

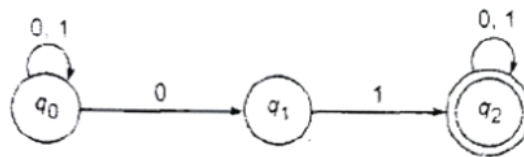
Time: 3 hours

Full Marks: 40

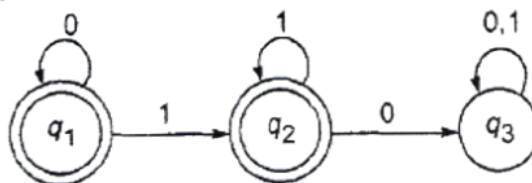
*Answer question number 1 and any **four** from the rest*

1. a) What are Mealy and Moore machines?
b) What do you mean by ambiguous grammar?
c) What is the advantage of PDA over FA?
d) State pumping lemma for regular sets. [2×4=8]

2. a) Considering the figure below, which of the following sentences are true?



- i) M is deterministic
 - ii) $\delta(q_1, 1)$ is defined
 - iii) 0101010101 is accepted by M
 - iv) 100110011001 is not accepted by M
 - v) $\delta(q_0, 01001) = \{q_1\}$
 - vi) $\delta(q_0, 011001) = \{q_0, q_1, q_2\}$
 - vii) $\delta(q_2, w) = q_2$ for any string $w \in \{0,1\}^*$
 - viii) $\delta(q_1, 11001) \neq \emptyset$
 - ix) $T(M) = \{w | w = x00y, \text{ where } x, y \in \{0,1\}^*\}$
 - x) A string having an odd number of 1's is accepted by M.
- b) With the help of an example explain how you can remove λ -moves. [0.5×10+3]
3. a) Construct a Mealy machine which can output EVEN, ODD according as the total number of 0's encountered is even or odd. The input symbols are 0 and 1.
b) Convert the Mealy machine obtained above into its equivalent Moore machine.
c) State Arden's theorem. What are its applications? [2+3+(2+1)]
 4. a) Find the regular expression accepted by the finite automaton for the transition diagram given below:



- b) Find the language generated by the grammar $S \rightarrow 0A \mid 1S \mid 0 \mid 1, A \rightarrow 1A \mid 1S \mid 1$
- c) Construct the context-free grammar to generate $\{0^m 1^n | 1 \leq m < n\}$. [4+2+2]

5. Consider a context-free grammar G with the following productions
 $S \rightarrow ASA \mid B, B \rightarrow aCb \mid bCa \mid C \rightarrow ACA \mid A, A \rightarrow a \mid b$

a) Find V_N and Σ .

b) Write three distinct strings of length 7 in $L(G)$.

c) Which of the following strings are in $L(G)$: aaa, bbb, aba, abb ?

d) Which of the following are true? $C \Rightarrow^* bab, C \Rightarrow^* bab, C \Rightarrow^* abab, C \Rightarrow^* AAA$

e) Check whether Λ is in $L(G)$ or not.

[1+2+2+2+1]

6. a) Construct the finite automata equivalent to the regular expression

$$(01 + 10)(0 + 1)^*(00+11)$$

b) Design a Turing machine over $\{0, b\}$ which can compute a concatenation function over $\Sigma = \{0\}$. If a pair of words $\{w_1, w_2\}$ is the input, the output has to be w_1w_2 . [4+4]

7. Write short notes on any two of the following :

a) Chomsky classification of languages

b) Equivalence of two finite automata

c) Removal of unit and null productions

[4×2=8]
